

# David Nesting

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I am an engineering leader, with an emphasis in complex systems, reliability, resiliency, security, and solving problems. My superpowers include working with empathy, communicating with non-technical leaders, and adapting. I enjoy edge cases, adversarial thinking, and being hands-on. My interests range from hacking electronics and AI to public policy.

## Qualifications

- Can effectively lead both incident response task forces and product delivery teams
- Can understand, build, and troubleshoot complex systems at all layers of the stack
- Can design and build agentic AI systems and AI orchestration platforms, and understand the security and reliability implications of these systems
- Can design and build complex systems with reliability, security, and privacy in mind
- Can write efficient and readable code (Go, Python, others)

TS/SCI

## Experience

### March 2025 - Present

- Artificial Intelligence and Software Defined Radio projects, public safety communications incident tracking, SDR orchestration on Kubernetes, AI agent workflow and orchestration. Primarily working in Go and Python.
- Wrote multiple Ubiquiti- and HikVision NVR-related wireshark dissectors to investigate a network intrusion
- Reliable and secure network design and deployment for a 30-unit residential building
- Guest Lecturer on Resiliency and Privacy Engineering for the Vanderbilt Policy Accelerator

### Expert Witness, Electronic Frontier Foundation (EFF), April-May 2025

- Consulted on IT modernization, cybersecurity, and privacy
- Testified as an Expert Witness at the US District Court for the Southern District of New York

### Vega-Bray Observatory Rehabilitation, March 2025

- Rehabilitated the Observatory's telescope, which had not seen light in 15 years
- Reverse engineered the dome control system and the telescope's drive and tracking system
- Built a custom hardware control system for the telescope's stepper motors, and wrote custom drive and tracking software in C/C++ (Arduino)

## **Office of the Federal CIO, The White House (OMB), 2024-2025**

### **AI and Cybersecurity Advisor to the Federal CIO**

- Artificial Intelligence Lead at OMB, building an AI program in collaboration with IT, policy, records, and legal teams
- Built on-premise AI platform proof of concept and associated tooling, using Go and Kubernetes with NVIDIA A100 hardware.
- AI, Cybersecurity, Reliability and Resiliency, and Technology Modernization policy and implementation advisor to the Federal CIO
- Technology Advisor to the OMB Chief Information Officer on AI, SRE/operations, data, cybersecurity, and modernization
- Led a cross-functional Technology Modernization Fund discovery effort and co-authored a \$~100M modernization proposal

## **Rebellion Defense, 2021-2023**

### **SRE and Infrastructure Team Manager**

- Ran the Site Reliability Engineering and Infrastructure team at Rebellion Defense, a startup building AI-powered defense software.
- 12 SREs and SWEs, mostly senior/staff level
- Platform was based on AWS, Kubernetes, and novel infrastructure
- This was a highly hands-on role, with ~50% of my work as an individual contributor

## **US Office of Personnel Management, 2019-2021**

### **Deputy Chief Information Officer (Term SES)**

- Managed a ~750-person federal CIO organization with a \$100M/year budget
- Built OPM's first Digital Service team, from PD design through hiring and onboarding.
- Led a mainframe modernization activity to improve OPM's resiliency against single points of failure
- Introduced modern collaboration tools, such as Zoom and Slack, to prepare the agency for extended COVID-19 telework
- Advised the CIO on security, risk, modern practices, resiliency, and engineering
- Led a deep dive into call center issues resulting in disastrous customer experience. Built a fully-functional, call-accepting reproduction of the call center using cloud tools in about 3 hours to disprove the belief that this was a multi-year level of effort
- One time I had to build a Prometheus-based monitoring system in Go in order to effectively troubleshoot a problem for which we had no visibility
- One time I had to reverse engineer the agency's correspondence tracking system because it finally died and nobody knew how it worked, wrote an entirely new stop-gap system in Go to allow users to access their data and extract records for archival

## **US Digital Service, The White House (OMB), 2014-2019**

### **Director of Engineering from 2017-2019:**

- Mentored a community of ~50 SREs, SWEs, and Data Engineers, supervising work assignments for 25

- Represented engineering and IT to agency and White House leadership, including to Cabinet officials
- Recruiting, speaking at round tables and conferences
- Co-designed the USDS engineering hiring pipeline

**Site Reliability and Security Engineer** from 2014-2019. Projects included:

- Multiple rapid response and modernization efforts, including security and availability incidents, at HHS/CMS, GSA, State, DOJ, and other agencies
- Database incident response with the Army
- Security Engineering, performing code reviews, advice on prioritization and mitigation.
- One time I wrote a Go-based ASP parser and data flow analyzer to automatically generate code fixes for thousands of vulnerabilities in a government system
- Various projects, researching and advising on security threats and capabilities of nation-state adversaries in classified and unclassified contexts
- Built and conducted bi-monthly security training and workshops for USDS
- Built and managed multiple internal tools in Python and Go, plus dozens of small software projects as part of my agency engagements

## Google

**Site Reliability Engineer** from 2007-2014 (7.5 years), on the logs infrastructure team, managing all layers of the logs infrastructure, including hardware, OS, and service, and supporting Google's legal and privacy teams. Projects included:

- On-call and incident response
- Full-stack support, from hardware through OS to service, for O(1M) CPU cores and exabytes of data
- Managed and maintained data integrity systems cryptographically proving the full data integrity of Google's logs infrastructure
- Built and maintained a data migration system for moving petabytes of data between clusters.
- Participated in Google's response to state-sponsored intrusions, leading to a major encryption and privilege reduction effort.
- SRE hiring and interviewing

## AT&T

**Technical Architect** and similar roles, from 1999-2007 (8 years), as a technical lead supporting [www.att.com](http://www.att.com) and similar sites. Introduced the DevOps model to the production operations team.

## Texas Networking, Inc.

**Engineer**, from 1995-1999 (approx.), as a member of the engineering staff. This was a small/startup regional ISP.

## Education

Texas A&M, Computer Engineering, 1998